DIESEL GENERATOR SET / 50HZ

WM-40 KVA

DIESEL GENERATOR SET

WATER-COOLED THREE-PHASE 50 HZ DIESEL

GENERATOR RATES

STANDBY POWER	PRIME POWER	SPEED	STANDARD VOLTAGE	POWER FACTOR
40 kVA	36 kVA	1500 rpm	400 / 230 V	0.8
32 kW	29 kW	50 Hz	Three-phase	Cos Phi

TECHNICAL SPECIFICATIONS

SPECIFICATIONS 50 HZ / 1500 RPM

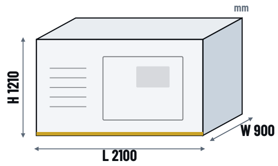
ENGINE SPECIFICATIONS — W-400D

Rated Output	32	kW
Manufacturer	WIRMAN	
Model	W-400D	
Engine Type	4 Stroke - Diesel	
Injection Type	Direct Injection	
Aspiration Type	Natural	
Number of cylinders and arrangement	4 In line	
Bore and Stroke	102 × 118	mm
Displacement	3.8	L
Cooling System	Water + 50 % antifreeze	
Lube Oil Specifications	ACEA E3,E4,E5 (15W-40)	
Compression Ratio	18:1	
Fuel Consumption Standby (110%)	11	l/h
Fuel Consumption 100% PRP	9	l/h
Fuel Consumption 75 % PRP	7	l/h
Fuel Consumption 50 % PRP	5	l/h
Lube oil consumption with full load	< 0.5	l/h
Total oil capacity including tubes, filters	13	L
Total coolant capacity	16.5	L
Governor	Mechanical	Type
Air Filter	Dry	Type

ALTERNATOR SPECIFICATIONS — SF184G

Manufacturer	WIRMAN	
Model	SF184G	
Poles	4	No.
Connection type (standard)	Star	
Total Harmonic Content	< 2 %	
Insulation	H Class	Class
Enclosure (according to IEC 34-5)	IP 21/23	
Exciter system	Self-excited, brushless	
Voltage regulator	A.V.R (Electronic)	
Bracket type	Single Bearing	
Coupling system	Flexible disc	
Efficiency (%)	87 %	
Voltage Output (VAC)	230 / 400	
Frequency (Hz)	50	

DIMENSIONS — CANOPIED



Weight	950	kg
Fuel Tank	100	lt

WxLxH **900 x 2100 x 1210** mm

SCOPE OF SUPPLY

STANDARD GENERATOR FEATURES

- Heavy duty water-cooled diesel engine and brushless alternator
- Solid steel base frame with minimum 12 hours base tank and anti-vibration mountings
- AMF control panel with digital automatic main control module
- 50 °C tropical radiator
- Battery charge rectifier, lead-acid battery
- Residential type silencer
- Flexible exhaust compensator
- Engine block water heater unit
- All rotating parts covered by metal mesh guards
- Engine and alternator manufacturer test reports (if provided)
- Factory load, performance and function tests
- User manual, electrical drawing and guarantee certificate

SOUNDPROOF CANOPY SPECIFICATIONS

- Special design for minimising acoustic level
- Galvanized steel construction further protected by polyester powder coat paint
- Black finish stainless steel locks and hinges
- Control panel viewing window in a lockable access door
- Emergency stop push button (red) mounted on enclosure exterior
- Lifting, drag and jacking points on base frame
- Radiator fill via removable, flush mounted rain cap fitted with compression seal

QUALITY & COMPLIANCE

WIRMAN has ISO 9001, ISO 14001 and OHSAS 18001 quality certifications.

ISO 9001 | ISO 14001 | OHSAS 18001 | CE

WIRMAN gensets are CE marked in accordance with the following directives:

EC DIRECTIVES

- 2006/42/EC Machinery safety
- 2014/30/EU Electromagnetic compatibility
- 2014/35/EU Electrical equipment designed for use within certain voltage limits
- 2000/14/EC Sound power level, noise emissions of outdoor equipment (amended by 2005/88/EC)
- 97/68/EC Emissions of gaseous and particulate pollutants (amended by 2002/88/EC & 2004/26/EC)
- EN 12100, EN 13857, EN 60204

POWER RATING DEFINITIONS

PRIME POWER (PRP):

Prime power is the maximum power which a generating set is capable of delivering continuously whilst supplying a variable electrical load when operated for an unlimited number of hours per year under the agreed operating conditions with the maintenance intervals and procedures being carried out as prescribed by the manufacturer. The permissible average power output (Ppp) over 24 h of operation shall not exceed 70 % of the PRP.

EMERGENCY STANDBY POWER (ESP):

Emergency standby power is the maximum power available during a variable electrical power sequence, under the stated operating conditions, for which a generating set is capable of delivering in the event of a utility power outage or under test conditions for up to 200 h of operation per year with the maintenance intervals and procedures being carried out as prescribed by the manufacturer. The permissible average power output over 24 h of operation shall not exceed 70 % of the ESP.

Note: All data based on operation to BS 5514 and DIN 6271 standard reference conditions.

CONTROL UNIT

DSE6120 MKIII

FRONT PANEL

1. Stop / Reset Mode
2. Manual Mode
3. Test Mode (DSE6120 MKIII only)
4. Auto Mode
5. Alarm Mute / Lamp Test
6. Start
7. Menu Navigation

KEY FEATURES

- 4-line back-lit LCD text display
- Multiple display languages
- Five-key menu navigation
- LCD alarm indication
- Customisable power-up text and screen images
- DSENet® expansion compatibility
- Data logging facility
- Internal PLC editor
- Protections disable feature
- Fully configurable via PC using USB communications
- Front panel configuration with PIN protection
- Power save mode
- 3-phase generator sensing and protection
- 3-phase mains (utility) sensing and protection (DSE6120 MKIII only)
- Automatic load transfer control (DSE6120 MKIII only)
- Generator current and power monitoring (kW, kvar, kVA, pf)
- Mains (utility) current and power monitoring (kW, kvar, kVA, pf) (DSE6120 MKIII only)
- kW overload alarm
- Over current protection
- Breaker control via fascia buttons
- Fuel and start outputs configurable when using CAN
- 6 configurable DC outputs
- 4 configurable analogue/digital inputs
- 8 configurable digital inputs
- CAN, MPU and alternator frequency speed sensing in one variant
- Real time clock
- Manual and automatic fuel pump control
- Engine pre-heat and post-heat functions
- Engine run-time scheduler
- Engine idle control for starting & stopping
- Fuel level alarms
- 3 configurable maintenance alarms
- Compatible with a wide range of CAN engines, including Tier 4 engine support
- Uses DSE Configuration Suite PC Software for simplified configuration
- Licence-free PC software
- IP65 rating (with optional gasket) offers increased resistance to water ingress
- Configurable CAN read & transmitted information
- 1 alternative configuration

KEY BENEFITS

- Automatically transfers between mains (utility) and generator (DSE6120 MKIII only) for convenience
- Hours counter provides accurate information for monitoring and maintenance periods
- User-friendly set-up and button layout for ease of use
- Multiple parameters are monitored & displayed simultaneously for full visibility
- The module can be configured to suit a wide range of applications for user flexibility
- PLC editor allows user configurable functions to meet user specific application requirements

RELATED MATERIALS

TITLE
DSE6110 MKIII & DSE6120 MKIII Installation Instructions
DSE6110 MKIII & DSE6120 MKIII Operator Manual
DSE6110 MKIII & DSE6120 MKIII Configuration Suite PC Manual